[INCH-POUND]
A-A-50193A
July 18, 1997
SUPERSEDING
A-A-50193
March 21, 1989

COMMERCIAL ITEM DESCRIPTION

# ETHYLENE ABSORBER/SHELF-LIFE EXTENDER, FOR FRESH FRUITS AND VEGETABLES

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers a porous fabric blanket containing ethylene absorbing material. The blanket is intended to extend the shelf-life of fresh fruits and vegetables when it is used in their storage.

2. SALIENT CHARACTERISTICS.

2.1 Absorbing materials. The absorbing material shall be activated alumina (aluminum oxide) coated or impregnated with potassium permanganate ($KMnO_4$) and shall have a maximum moisture content of 16 percent (%). The absorbing material shall be spherical or pelletized in shape and shall be porous for efficient removal of ethylene from the atmosphere. The absorbing material shall be bright purple in color and free of dust.

2.2 Construction. The blanket shall contain a range of 10 to 15 pounds (lb) of absorbing material. The fabric shall be a porous barrier such as non-woven polyester or similar material which allows passage of air, and is dense enough to prevent pellets or excessive particle fines from escaping to the atmosphere. The blanket shall be constructed to withstand tearing during normal placement in refrigerated walk-in boxes or shipping containers. The blanket shall be rectangular in shape and the surface area shall be in the range of 500 to 1200 square inches ($in^2$) per side. Eyelets to facilitate hanging shall be built into the outer borders of the blanket. Eyelets shall be placed so that the absorbing material will distribute evenly and not be compacted to one end when in a hanging position. A tag shall be affixed to the blanket and shall have the following statement: "AVOID CONTACT WITH WATER".

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Defense Logistic Agency, Defense Supply Center Richmond, ATTN: DSCR-VBD, Richmond, VA 23297-5610.

AMSC N/A FSC 6850

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2.3 Performance.

2.3.1 Moisture analysis. Analysis for moisture shall be made in accordance with the latest edition of the Official Methods of Analysis of the Association of Official Analytical Chemists:

| Test | Source | | Method |
|---|---|---|---|
| Moisture | Chapter: | Cereal foods | Air oven method |
| | Section: | Wheat flour | |

2.3.2 Potassium permanganate. The absorbing material shall have a minimum content of 4 % $KMnO_4$ by weight to provide adequate capacity for ethylene removal. The determination for weight percentage of $KMnO_4$ shall be as follows:

a. Prepare a standard $KMnO_4$ solution by mixing 0.100 gram (g) $KMnO_4$ in 100 milliliters (mL) of water and then dilute it 20-fold (1 volume of $KMnO_4$ solution plus 19 volumes of water).
b. Read the absorbency of the $KMnO_4$ solution on a spectrophotometer at 525 nanometers against water as the blank ($A_{std}$).
c. With a mortar and pestle, pulverize the absorbing material into a fine powder. Mix 1 g of the powder with 100 mL of water in a flask and stir gently for 10 minutes. Filter an aliquot of the solution.
d. Dilute the absorbing material solution 10-fold and read the absorbency at 525 nanometers ($A_{sample}$).
e. Determine percentage using the formula:

$$\% KMnO_4 = (A_{sample} / A_{std}) \times 100/20$$

3. QUALITY ASSURANCE PROVISIONS.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, examinations, and tests.

3.2 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 5.2).



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5. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5 1.1 Non-Government documents. Copies of the Official Methods of Analysis of the Association of Official Analysis Chemists are available from the Association of Official Analytical Chemists, 481 North Fredrick Avenue, Suite 500, Gaithersburg, MD 20877-2417

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Selection of applicable level and packaging requirements.

MILITARY INTERESTS:

Custodians:
Army - GL1
Navy - MC
Air Force - 68
DLA - GS

Review Activities:
Army - MD1
Navy - SH
DLA - SS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
DLA - GS

(Project 6850-1208)



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